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Lifter assembly with bellows for optical inspection system

Abstract

A lifter assembly for an optical system includes a chuck having a chuck base and a detachable top plate, wherein the detachable top plate is one of a plurality of interchangeable detachable top plates configured to support a substrate. The lifter assembly also includes a moveable plate supporting the chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position. The lifter assembly further includes a bellows structure operatively coupled to the moveable plate. The lifter assembly yet further includes a pneumatic system fluidly coupled to the bellows structure to selectively expand and contract, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Application No. 63/430,412, filed Dec. 6, 2022, the disclosure of which is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) The embodiments described herein relate to optical inspection systems and, more particularly, to a pneumatic lifter assembly for such systems.

BACKGROUND

(2) Inspection processes are used at various steps during a semiconductor manufacturing process to detect defects on wafers to promote higher yield in the manufacturing process, thereby obtaining higher profits. Inspection has always been an important part of fabricating semiconductor devices. However, as the dimensions of semiconductor devices decrease, inspection becomes even more important to the successful manufacture of acceptable semiconductor devices since smaller defects can cause the devices to fail.

(3) Some fabrication, metrology and defect inspection processes require vertical movement of a chuck on which a substrate is positioned on a top surface of A lifter assembly supports the chuck and actuates the vertical movement. For example, a pneumatic rod may be located within a chamber of the lifter assembly and connected to support features and a movable plate to carry out the vertical movement of the chuck with the introduction and removal of a fluid. Some tools that the lifter assembly works in conjunction with may offer limited space to accommodate the overall lifter assembly features during operation. The aforementioned pneumatic rod occupies more space when the lifter assembly is extended away from a retracted (i.e., down) position to an upward position. Furthermore, the pneumatic rod may not be able to maintain perpendicularity to the support features it is adjacent to when an uneven load is being lifted or applied, thereby resulting in wear inside the bore within which the rod slides. Finally, a pneumatic rod is only capable of extending to a fixed stroke length that is not adjustable.

(4) The above-described issues pose challenges to the inspection process of substrates.

SUMMARY OF THE DISCLOSURE

(5) According to one aspect of the disclosure, a lifter assembly for an optical system includes a chuck having a chuck base and a detachable top plate, wherein the detachable top plate is one of a plurality of interchangeable detachable top plates configured to support a substrate. The lifter assembly also includes a moveable plate supporting the chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position. The lifter assembly further includes a bellows structure operatively coupled to the moveable plate. The lifter assembly yet further includes a pneumatic system fluidly coupled to the bellows structure to selectively expand and contract, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position.

(6) According to another aspect of the disclosure, an optical system includes a substrate chucking sub-system having a chuck having a chuck base and a detachable top plate, the detachable top plate configured to support a substrate. The optical system also includes a moveable plate supporting the chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position. The optical system further includes a bellows structure operatively coupled to the moveable plate, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position. The optical system yet further includes an illumination source configured to illuminate one or more portions of the substrate. The optical system also includes a detector configured to collect illumination from the illuminated one or more portions of the substrate.

(7) According to yet another aspect of the disclosure, a lifter assembly for vertically repositioning a

substrate includes a vacuum chuck configured to support the substrate and hold the substrate with vacuum air pressure. The lifter assembly also includes a moveable plate supporting the vacuum chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position. The lifter assembly further includes a welded bellows structure operatively coupled to the moveable plate, wherein expansion of the welded bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position.

(8) These aspects and other advantages and features are apparent from the following description taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter that is regarded as embodiments of the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The features and advantages of the disclosed embodiments are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a partial cross-sectional, elevation view of an optical inspection system lifter assembly supporting a chuck;

(3) FIG. 2 is a perspective view of the lifter assembly;

(4) FIG. 3A illustrates the lifter assembly in a retracted position with fluid lines connected thereto;

(5) FIG. 3B illustrates the lifter assembly in an extended position with fluid lines connected thereto;

(6) FIG. 4A illustrates the lifter assembly in the retracted position;

(7) FIG. 4B illustrates the lifter assembly in the extended position;

(8) FIG. 5 is a cross-sectional, elevation view of the lifter assembly according to another aspect of the disclosure;

(9) FIG. 6A schematically illustrates an optical system incorporating the embodiments of the lifter assembly disclosed herein according to one aspect of the disclosure; and

(10) FIG. 6B schematically illustrates an optical system incorporating the embodiments of the lifter assembly disclosed herein according to another aspect of the disclosure.

DETAILED DESCRIPTION

(11) Referring to the Figures, where the invention will be described with reference to specific embodiments, it is to be understood that the disclosed embodiments are merely examples of the invention that may be embodied in various and alternative forms. The Drawings are not necessarily to scale, as some features may be exaggerated, reduced or simplified to show details of particular components. Therefore, specific structural and functional details disclosed herein and illustrated in the Drawings are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the invention.

(12) Referring to FIG. 1, a portion of a substrate chucking system for an optical system is shown and generally referenced with numeral **11**. Embodiments of a lifter assembly **10** are disclosed herein to support and vertically adjust a chuck **14**. The lifter assembly **10** includes a moveable plate **12** which supports the chuck **14**. The chuck **14** is configured to support a substrate, such as a semiconductor wafer (not shown), for example. Various types of contemplated chucks may be used with the lifter assembly **10** disclosed herein. By way of non-limiting example, the lifter assembly **10** may be used with the substrate chucking system disclosed in U.S. patent application Ser. No. 17/562,544, the disclosure of which is incorporated by reference herein in its entirety. In the illustrated embodiment, the chuck **14** is a vacuum chuck that holds the substrate in place on a top

surface **13** of the chuck **14** with vacuum air pressure routed through the chuck **14** via one or more vacuum inlets **15** and vacuum reservoirs **17**. Additionally, in the illustrated embodiment, the chuck **14** includes a chuck base **16** and a detachable plate **18**. The detachable plate **18** is secured to the chuck base **16** with one or more fasteners or guides **20**. However, it is to be understood that it is within the scope of the disclosed embodiments to have a single structure chuck **14**.

(13) The chuck base **16** is structured to advantageously allow different types of the detachable plates **18** to be secured thereon. The detachable plate **18** is suitable for quick interchange on the chuck base **16**. Such a configuration is more time and cost efficient in the substrate chuck interchanging process, as it allows for a common chuck base to accommodate multiple top plates used for a variety of substrates.

(14) Regardless of the specific type of chuck, overall tool/stage and optical system, the embodiments of the lifter assembly **10** disclosed herein include a bellows structure **22** that actuates substantially vertical movement of the moveable plate **12** and therefore the chuck **14** and substrate supported thereon. The bellows structure **22** is a welded bellows type in some embodiments. The bellows structure **22** is formed in an accordion structure so that a plurality of annular bellows plates having radially curved surfaces are connected in an alternating manner on the outside-diameter side and the inside-diameter side. The bellows plates are connected together by TIG welding or the like in the atmosphere or an Ar gas environment, for example. The bellows structure **22** defines a hollow interior chamber.

(15) Vertical adjustment of the moveable plate **12** alters a distance between the chuck **14**—with the substrate held thereon—and a measurement device of the optical system. As used herein, the term “measurement device” is generally used to refer to a metrology device, an inspection device, or a combination metrology and inspection device. Vertical adjustment of the moveable plate **12**, chuck **14** and substrate facilitates various actions by the optical system.

(16) Referring now to FIG. 2, the lifter assembly **10** is shown in additional detail. As shown, the fasteners and guides **20** are radially outward of the vacuum inlets **15** in the illustrated embodiment, which is the opposite configuration to that shown in FIG. 1. Accordingly, it is to be appreciated that either configuration is contemplated.

(17) The moveable plate **12** in the illustrated embodiment has a central portion **24** and a plurality of arms **26** extending from the central portion **24**. In particular, the illustrated embodiment has three arms, but it is to be appreciated that alternative numbers of arms may be present in other embodiments. The lifter assembly **10** also includes a base flange **28** and a top cap **30**. A plurality of guiding shafts **32** each extend through respective holes of the moveable plate **12** and are operatively coupled to the base flange **28** and the top cap **30**. As shown in FIGS. 2-4B, the guiding shafts **32** are located external to, or radially outward of, the bellows structure **22**. Each guiding shaft **32** may have threaded ends to engage threaded regions of the base flange **28** and top cap **30**. Alternatively, the guiding shafts **32** may be coupled to the base flange **28** and the top cap **30** with nuts or the like. While the bellows structure **22** actuates vertical adjustment of the moveable plate **12**, the chuck **14** and the substrate, the guiding shafts **32** guide the vertical movement to maintain adjustment in the vertical direction. The number of the plurality of guiding shafts **32** may vary depending upon the particular application of use.

(18) Referring now to FIGS. 3A and 3B, with continued reference to FIG. 2, the lifter assembly **10** is fluidly coupled to a pneumatic system **40**. A first fluid connection is made between at least one of the plurality of arms **26** of the moveable plate **12** via a first piping arrangement **42**. The first fluid connection provides the vacuum air pressure required to hold the substrate on the chuck **14**, as described above. The second fluid connection is made between an interior chamber of the bellows structure **22** and a selectively switchable supply from a compressed air source **44** and a vacuum air pressure source **46** via a second piping arrangement **48**. Regarding the second fluid connection, a ported solenoid **50** has an inlet **52** to receive compressed air from the compressed air source **44**, an outlet **54** to route the compressed air to the interior chamber of the bellows structure **22**, and an

exhaust **56** to withdraw compressed air from the interior chamber of the bellows structure **22** with the vacuum air pressure source **46**.

(19) The ported solenoid **50** in the illustrated embodiment is a 3-way solenoid, but it is to be appreciated that other types of piping arrangements may be utilized for the second fluid connection to selectively route compressed air into and out of the interior chamber of the bellows structure **22**. In the illustrated embodiment of the ported solenoid **50**, vacuum air pressure is provided to the bellows structure **22** to contract the bellows structure **22**, thereby lowering the moveable plate **12**, when the ported solenoid **50** is de-energized. Conversely, upon energization of the ported solenoid **50**, compressed air enters the bellows structure **22** through a pressure regulator **58** that is pre-set to a pressure value sufficient to expand the bellows structure **22**, thereby lifting the moving mass which includes the moveable plate **12**, the chuck **14** and the substrate supported thereon.

(20) Referring to FIGS. **4A** and **4B**, the lifter assembly **10** according to a first embodiment is illustrated in greater detail. The bellows structure **22** is vertically adjustable between a lower, retracted position (FIG. **4A**) and an upper, extended position (FIG. **4B**) to vertically adjust the moveable plate **12**, the chuck **14** and the substrate. As described above, the plurality of guiding shafts **32** extend through respective holes of the moveable plate **12** and are operatively coupled to the base flange **28** and the top cap **30** to guide vertical movement of the moveable plate **12**. Stabilization of movement of the moveable plate **12** during vertical translation along the guiding shafts **32** is provided with a plurality of bushing arrangements **60**. In particular, the moveable plate **12** has a number of bushing arrangements **60** mounted on an upper surface **62** thereof which corresponds to the number of guiding shafts **32**. Therefore, each guiding shaft **32** passes through a respective bushing arrangement **60**. Each bushing arrangement **60** includes a bush housing **64** which contains a bushing **68** therein. The bushing arrangements **60** concentrically surround a portion of the guiding shafts **32** and translate with the moveable plate **12**.

(21) The lower, retracted position (FIG. **4A**) and the upper, extended position (FIG. **4B**) of the lifter assembly **10** are defined by respective upper and lower stop structures, as described herein. The disclosed stop structures provide pre-defined limits of vertical travel for the bellows structure **22**, the moveable plate **12**, the chuck **14** and therefore the substrate held thereon.

(22) The upper, extended position is defined by contact between an upper travel limit feature **70** and an upper stopper **72** operatively coupled to, or integrally formed with, the moveable plate **12**. In some embodiments, the upper travel limit feature **70** is integrally formed with the top cap **30** and is part of a lower surface **74** of the top cap **30**. In other embodiments, the upper travel limit feature **70** is a component separate from the top cap **30** and is operatively coupled thereto, as illustrated. For example, the upper travel limit feature **70** may be directly threaded to the top cap **30** or coupled thereto with a mechanical fastener. The upper travel limit feature **70** is positioned to be contacted by the upper stopper **72** which extends upwardly from the upper surface **62** of the moveable plate **12**. As noted above, the upper stopper **72** may be integrally formed with the moveable plate **12**. In embodiments with the upper stopper **72** operatively coupled to the moveable plate **12**, the upper stopper **72** may have a coupling segment **76** extending through the moveable plate **12** for threading to the moveable plate **12** or may completely extend through the moveable plate **12** to be coupled to an upper flange **78** disposed below the moveable plate **12**. Any of the embodiments disclosed herein results in contact between the upper travel limit feature **70** and the upper stopper **72** to define the upper, extended position of the moveable plate **12**.

(23) The lower, retracted position is defined by contact between a lower travel limit feature **80** and a lower stopper **82** operatively coupled to, or integrally formed with, the moveable plate **12**. In the illustrated embodiment, the upper flange **78** includes a main body portion **84** disposed between the top of the bellows structure **22** and the moveable plate **12**. The upper flange **78** also includes a downwardly protruding portion which is the lower stopper **82** in the illustrated embodiment. The lower stopper **82** is positioned to contact an upwardly protruding portion of the base flange **28** which is the lower travel limit feature **80** in the illustrated embodiment. Contact between the lower

stopper **82** and the lower travel limit feature **80** define the lower, retracted position of the moveable plate **12**. It is to be appreciated that in alternative embodiments, the lower travel limit feature **80** may be a component which is operatively coupled to the base flange **28** rather than integrally formed therewith. Similarly, the lower stopper **82** may be operatively coupled to the upper flange **78** or the moveable plate **12** in other embodiments.

(24) Referring now to FIG. 5, a slider assembly **90** is provided as an alternative to the plurality of guiding shafts **32** and bushing arrangements **60** shown in FIGS. 4A and 4B. The slider assembly **90** includes a plurality of slide rails **92** extending through respective holes defined by the moveable plate **12**. Each of the slide rails **92** is operatively coupled to the base flange **28** and the top cap **30**. Each sliding rail **92** may have threaded ends to engage threaded regions of the base flange **28** and top cap **30**. Alternatively, the sliding rails **92** may be coupled to the base flange **28** and the top cap **30** with nuts or the like. While the bellows structure **22** actuates vertical adjustment of the moveable plate **12**, the chuck **14** and the substrate, the sliding rails **92** guide the vertical movement to maintain adjustment in the vertical direction. The number of the plurality of slide rails **92** may vary depending upon the particular application of use.

(25) A plurality of mount plates **94** corresponding to the number of the plurality of slide rails **92** are coupled to the upper surface **62** of the moveable plate **12**. Each of the mount plates **94** have a sliding member **96** disposed between the respective mount plate **94** and slide rail **92**. Accordingly, during vertical movement of the moveable plate **12**, the slider assembly **90** ensures smooth, guided movement of the moveable plate **12**.

(26) Referring now to FIGS. 6A and 6B, simplified schematic views of optical systems **200**, **210** equipped with the lifter assembly **10** and the chuck **14**, as described previously herein are illustrated. In one embodiment, the optical systems **200**, **210** further include at least one light source **202** (e.g., one or more lasers, one or more broadband light sources, etc.) configured to illuminate an area on the surface of the substrate. In another embodiment, the optical systems **200**, **210** include one or more detectors **204**, or cameras, suitable for detecting light reflected, diffracted, or scattered from the area illuminated by the light source **202**. In one embodiment, the one or more detectors **204** may include, but are not limited to, a CCD or TDI-CCD detector, or a photomultiplier detector. In addition, the optical systems **200**, **210** may include a set of optical elements (e.g., illumination optics, collection optics, beam splitters **206**, filters and the like) configured for directing (and focusing) illumination from the light source **202** onto the surface of the substrate and, in turn, directing illumination from the surface of the substrate to the imaging portion of the detector **204** of the optical systems **200**, **210**. For instance, the set of optical elements for the optical systems **200**, **210** may include, but is not limited to, a primary imaging lens suitable for imaging the illuminated area of the substrate onto a collection portion(s) of the detector **204**. Further, the imaging detector **204** may be communicatively coupled to an image processing computer which may identify and store imagery data acquired from the detector **204**.

(27) The optical systems **200**, **210** disclosed herein may be configured as any optical systems known in the art, including, but not limited to inspection systems, metrology systems, and lithography systems. For example, as shown in FIG. 6A, the optical system **200** may be configured as a bright field (BF) inspection system. Alternatively, as shown in FIG. 6B, the optical system **210** may be configured as a dark field (DF) inspection system. It is noted that the optical configurations depicted in FIGS. 6A and 6B are provided merely for illustrative purposes and should not be interpreted as limiting. In a general sense, the optical systems **200**, **210** may include any set of imaging and optical elements suitable for imaging the surface of the substrate.

(28) The embodiments disclosed herein provide a lifter assembly **10** that is more compact, relative to other lifting devices. Accordingly, the lifter assembly **10** can fit in smaller overall spaces and may be configured to be used in different optical systems. Additionally, wear and tear on the lifter assembly components is reduced due to the frictionless actuation provided by the bellows structure **22**, as this actuation accommodates tilted, cylinder rod rotation and uneven loads.

(29) While the invention has been described in detail in connection with only a limited number of embodiments, it is to be readily understood that the invention is not limited to such disclosed embodiments. Rather, the invention can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the invention. Additionally, while various embodiments of the invention have been described, it is to be understood that aspects of the invention may include only some of the described embodiments. Moreover, any feature, element, component or advantage of any one embodiment can be used on any of the other embodiments. Accordingly, the invention is not to be seen as limited by the foregoing description.

Claims

1. A lifter assembly for an optical system, the lifter assembly comprising: a chuck comprising a chuck base and a detachable top plate, wherein the detachable top plate is one of a plurality of interchangeable detachable top plates configured to support a substrate; a moveable plate supporting the chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position; a bellows structure operatively coupled to the moveable plate; a plurality of guiding shafts extending through holes of the moveable plate to guide vertical adjustment of the moveable plate, the plurality of guiding shafts located external to the bellows structure; and a pneumatic system fluidly coupled to the bellows structure to selectively expand and contract, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position.
2. The lifter assembly of claim 1, wherein the pneumatic system comprises a solenoid having an inlet port, an outlet port, and an exhaust port, the inlet port fluidly coupled to a compressed air source to selectively provide compressed air through the outlet port and into an interior chamber of the bellows structure to expand the bellows structure, the exhaust port fluidly coupled to a vacuum generating device to withdraw compressed air from the interior chamber of the bellows structure to contract the bellows structure.
3. The lifter assembly of claim 1, further comprising: a top cap; a base flange; and wherein each of the plurality of guiding shafts extending through holes of the moveable plate to guide vertical adjustment of the moveable plate, each of the guiding shafts is operatively coupled to the top cap and to the base flange.
4. The lifter assembly of claim 3, further comprising a stopper projecting upwardly from the upper surface of the moveable plate.
5. The lifter assembly of claim 4, further comprising an upper travel limit feature integrally formed with the top cap, wherein contact between the upper travel limit feature and the stopper operatively define the upper, extended position of the moveable plate.
6. The lifter assembly of claim 4, further comprising an upper travel limit feature operatively coupled to the top cap, wherein contact between the upper travel limit feature and the stopper operatively define the upper, extended position of the moveable plate.
7. The lifter assembly of claim 3, further comprising an upper flange having a main body portion disposed between a top surface of the bellows structure and a lower surface of the moveable plate, the upper flange also having a downwardly protruding portion positioned to contact an upwardly protruding portion of the base flange to define the lower, retracted position of the moveable plate.
8. The lifter assembly of claim 1, further comprising a plurality of bush housings corresponding to the number of the plurality of guiding shafts, each of the bush housings coupled to the moveable plate and having a bushing disposed therein, each bushing concentrically surrounding a respective one of the plurality of guiding shafts.
9. The lifter assembly of claim 1, wherein the bellows structure is a welded bellows structure.

10. The lifter assembly of claim 1, wherein the substrate comprises a semiconductor wafer.
 11. An optical system comprising: a substrate chucking sub-system having: a chuck having a chuck base and a detachable top plate, the detachable top plate configured to support a substrate; a moveable plate supporting the chuck on an upper surface of the moveable plate, the moveable plate vertically adjustable between a lower, retracted position and an upper, extended position; a bellows structure operatively coupled to the moveable plate, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position; and a plurality of guiding shafts extending through holes of the moveable plate to guide vertical adjustment of the moveable plate, the plurality of guiding shafts located external to the bellows structure; an illumination source configured to illuminate one or more portions of the substrate; and a detector configured to collect illumination from the illuminated one or more portions of the substrate.
 12. The optical system of claim 11, wherein the optical system is configured as an inspection tool.
 13. The optical system of claim 11, wherein the optical system is configured as a metrology tool.
 14. The optical system of claim 11, further comprising a pneumatic system fluidly coupled to the bellows structure to selectively expand and contract, wherein expansion of the bellows structure vertically adjusts the moveable plate to the upper, extended position and contraction of the bellows structure vertically adjusts the moveable plate to the lower, retracted position.
 15. The optical system of claim 14, wherein the pneumatic system comprises a solenoid having an inlet port, an outlet port, and an exhaust port, the inlet port fluidly coupled to a compressed air source to selectively provide compressed air through the outlet port and into an interior chamber of the bellows structure to expand the bellow structure, the exhaust port fluidly coupled to a vacuum generating device to withdraw compressed air from the interior chamber of the bellows structure to contract the bellows structure.
 16. The optical system of claim 11, further comprising: a top cap; a base flange; and wherein each of the plurality of guiding shafts extending through holes of the moveable plate to guide vertical adjustment of the moveable plate, each of the guiding shafts is operatively coupled to the top cap and to the base flange.
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